

ECON 381: Practice Midterm 1 — Answers

Professor MUYANG REN

Fall 2026

1. Multiple Choice Answers (30 points; 3 each)

Question	1	2	3	4	5	6	7	8	9	10
Answer	c	b	d	a	c	b	d	a	c	b

1. **(c)** Each observation is one school district.
2. **(b)** The fitted change is $(-2.28)(-2) = 4.56$ test-score points.
3. **(d)** The population error collects influences on test scores not represented by STR_i in the population line.
4. **(a)** The population mean is fixed, while the sample mean generally changes across random samples.
5. **(c)** This conditional expectation averages test scores within the population of districts in Los Angeles County.
6. **(b)** Correlation is unchanged by converting income to dollars. The positive scaling factor cancels between covariance and the standard deviation of income.
7. **(d)** Both intervals are centered at $\bar{Y}$. The 95% interval uses a larger critical value and is wider (for a positive standard error).
8. **(a)** OLS minimizes the sum of squared residuals, $\sum_i [Y_i - (b_0 + b_1 X_i)]^2$.
9. **(c)** Since $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, the fitted line passes through $(\bar{X}, \bar{Y})$.
10. **(b)** The SER measures the typical residual size in the units of Y .

2. True or False Answers (20 points; 5 each)

- (a) **False.** The Law of Large Numbers says that, under the conditions discussed in class, the probability of a large discrepancy between $\bar{Y}$ and μ_Y becomes small as the sample grows. It does not guarantee a monotonic approach: adding an observation can move the realized mean farther from the population mean.
- (b) **True.** The sample standard deviation s_Y measures dispersion among individual observations. For an i.i.d. sample, $SE(\bar{Y}) = s_Y/\sqrt{n}$ estimates the standard deviation of sample means across repeated samples of size n .
- (c) **False.** The confidence interval concerns the population mean, not individual outcomes. When the normal approximation is accurate, approximately 95% of intervals constructed in repeated random samples contain the fixed population mean.
- (d) **True.** Covariance scales linearly when one variable is multiplied by a constant:

$$\begin{aligned}\text{Cov}(2X, Y) &= E[(2X - E[2X])(Y - E[Y])] \\ &= 2E[(X - E[X])(Y - E[Y])] \\ &= 2\text{Cov}(X, Y).\end{aligned}$$

This is the covariance scaling property discussed in Handout 4.

3. Analytical Answers (50 points)

The observations are $(1, 3), (2, 3), (3, 6)$, with $n = 3$.

- (a) **Sample means (4 points; 2 each).**

$$\bar{X} = \frac{1 + 2 + 3}{3} = 2, \quad \bar{Y} = \frac{3 + 3 + 6}{3} = 4.$$

- (b) **Sample standard deviation (5 points).** The deviations in Y are $-1, -1, 2$. Therefore,

$$s_Y^2 = \frac{(3 - 4)^2 + (3 - 4)^2 + (6 - 4)^2}{3 - 1} = \frac{1 + 1 + 4}{2} = 3, \quad s_Y = \sqrt{3}.$$

- (c) **Standard error of the mean (4 points).**

$$SE(\bar{Y}) = \frac{s_Y}{\sqrt{n}} = \frac{\sqrt{3}}{\sqrt{3}} = 1.$$

- (d) **Confidence intervals (8 points; 4 each).** The normal-approximation formula is $\bar{Y} \pm cSE(\bar{Y})$. Rounding the critical values displayed in Handout 4, 1.645 and 1.960, to two decimal places gives 1.65 and 1.96:

$$CI_{90\%} = 4 \pm 1.65(1) = [2.35, 5.65],$$
$$CI_{95\%} = 4 \pm 1.96(1) = [2.04, 5.96].$$

Rounding note: The unrounded 95th standard normal percentile is approximately 1.64485, which rounds directly to 1.64. Thus $[2.36, 5.64]$ is also an acceptable 90% interval. Accept either convention when used consistently; the difference arises from rounding the handout's already-rounded value.

- (e) **Coverage (4 points). No.** With only three observations, the CLT-based normal approximation may be poor. The procedure is not guaranteed to cover μ_Y in 95% of repeated samples. A nominal 95% confidence level describes the repeated-sampling procedure when its approximation is accurate.

Analytical Answers (continued)

- (f) **OLS slope (10 points).** The deviations from the sample means are

$$X_i - \bar{X} = (-1, 0, 1), \quad Y_i - \bar{Y} = (-1, -1, 2).$$

Hence,

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{(-1)(-1) + (0)(-1) + (1)(2)}{(-1)^2 + 0^2 + 1^2} = \boxed{\frac{3}{2}}.$$

The numerator is 3 and the denominator is 2.

- (g) **Intercept and fitted equation (6 points).**

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 4 - \frac{3}{2}(2) = 1}, \quad \boxed{\hat{Y}_i = 1 + \frac{3}{2}X_i}.$$

- (h) **Fitted values and residuals (6 points; 1 per entry).** Use $\hat{Y}_i = 1 + 1.5X_i$ and $\hat{u}_i = Y_i - \hat{Y}_i$:

i	X_i	Y_i	$\hat{Y}_i$	$\hat{u}_i$
1	1	3	2.5	0.5
2	2	3	4	-1
3	3	6	5.5	0.5

As a check, the residuals sum to $0.5 - 1 + 0.5 = 0$.

- (i) **R^2 (3 points).**

$$TSS = \sum_i (Y_i - \bar{Y})^2 = 1 + 1 + 4 = 6,$$

$$RSS = \sum_i \hat{u}_i^2 = (0.5)^2 + (-1)^2 + (0.5)^2 = 1.5,$$

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{1.5}{6} = \boxed{0.75}.$$

The fitted line captures 75% of the total sample variation in Y .

4. Bonus Answers (5 points)

Here $X_i = 2$ for every observation, $\bar{X} = 2$, and $\bar{Y} = 4$.

- (a) **Well-defined slope (3 points). No.** Every $X_i - \bar{X}$ is zero. Both the numerator and denominator of the usual slope formula equal zero:

$$\frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{0}{0}.$$

This is undefined, not a slope of zero. There is no variation in X with which to estimate a unique slope when the intercept is also present.

- (b) **Unique slope when the intercept is fixed (2 points). Yes.** Since every $X_i = 2$, all fitted values equal the same constant $b_0 + 2\hat{\beta}_1$. The the sample-mean relationship discussed in class gives

$$\bar{Y} = b_0 + \hat{\beta}_1 \bar{X}.$$

Substituting $\bar{Y} = 4$ and $\bar{X} = 2$,

$$4 = b_0 + 2\hat{\beta}_1 \quad \implies \quad \boxed{\hat{\beta}_1 = \frac{4 - b_0}{2}}.$$

For any specified b_0 , this equation gives exactly one slope because $\bar{X} = 2 \neq 0$.